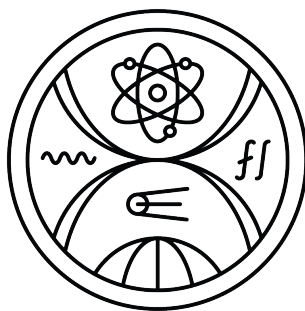


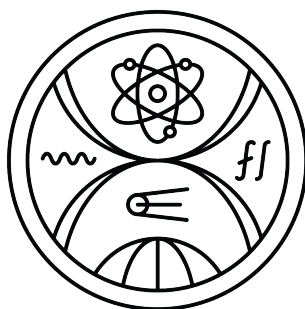
COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



**BENCHMARKING MATHEMATICAL
REASONING BASED ON VISUAL INPUT IN
LARGE LANGUAGE MODELS IN SLOVAK**

Master thesis

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Master thesis

Study program: Applied informatics
Branch of study: Applied informatics
Department: Department of Applied Informatics
Supervisor: Mgr. Marek Šuppa



Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

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Študijný odbor: informatika
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Jazyk záverečnej práce: anglický
Sekundárny jazyk: slovenský

Názov: Benchmarking Mathematical Reasoning Based on Visual Input in Large Language Models in Slovak

Porovnávacie hodnotenie matematického uvažovania na základe vizuálneho vstupu vo veľkých jazykových modeloch v slovenčine

Anotácia: Nedávne pokroky v oblasti veľkých jazykových modelov (LLM) ukázali ich potenciál pri riešení zložitých úloh zameraných na odôvodňovanie, ktoré sa spoliehajú na vizuálny vstup pre riešenie matematických problémov. Tieto úlohy sa často hodnotia pomocou štandardizovaných testov, ako sú olympiády alebo maturitné skúšky, ktoré vyžadujú nielen počítacie schopnosti, ale aj schopnosť interpretovať diagramy, grafy a vizuálne podnety. Napriek tomu pretrvávajú určité obmedzenia: údaje zo štandardizovaných testov sa často prekrývajú s tréningovými súdbami LLM, čo vedie k skresleným hodnoteniam. Väčšina benchmarkov sa navyše zameriava na angličtinu, pričom v slovenčine, ktorá je jazyk s nízkymi zdrojmi, je dostupných len málo zdrojov. Tento nedostatok venovaných benchmarkov bráni hodnoteniu skutočných schopností LLM v slovenčine a ich použiteľnosti v rôznych vzdelávacích kontextoch.

Cieľ: Ciele práce zahŕňajú (ale nie sú obmedzené na):

- Preskúmanie, ako LLM pristupujú k matematickému odôvodňovaniu založenému na vizuálnych vstupoch.
- Skúmanie existujúcich benchmarkov na hodnotenie vizuálneho a matematického odôvodňovania.
- Identifikáciu a riešenie medzier v benchmarkoch pre slovenčinu.
- Vypracovanie slovenského datasetu matematických úloh s vizuálnymi komponentmi zo štandardizovaných testov a olympiád.
- Hodnotenie výkonu LLM na datasetoch so zameraním na odôvodňovanie prostredníctvom vizuálnych vstupov.
- Analýzu chýb v odôvodňovaní založenom na vizuálnych úlohách v slovenskom kontexte.

Literatúra: Shi, Freda, et al. "Language models are multilingual chain-of-thought reasoners." arXiv preprint arXiv:2210.03057 (2022). (<https://arxiv.org/abs/2210.03057>)
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Chen, Nuo et al. "Breaking Language Barriers in Multilingual Mathematical Reasoning: Insights and Observations." ArXiv abs/2310.20246 (2023): n. (<https://arxiv.org/abs/2310.20246>)

Lu, Pan et al. "MathVista: Evaluating Mathematical Reasoning of Foundation Models in Visual Contexts." International Conference on Learning Representations (2023).

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Dátum zadania: 14.12.2024

Dátum schválenia: 15.12.2024

prof. RNDr. Roman Ďuríkovič, PhD.
garant študijného programu

.....
študent

.....
vedúci práce

I hereby declare that I have written this thesis by myself, only with help of referenced literature, under the careful supervision of my thesis advisor.

Bratislava, 2026

.....
Lea Gogová

Acknowledgement

Abstract

tba

Keywords:

Abstrakt

tba

Klíčové slová:

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Motivation

tba

Chapter 1

Vision-language models

Chapter 2

Reasoning

[Should define what we mean by *reasoning tasks* and when do we say that models *reason*.]

2.1 Benchmarks

One of the early large-scale math datasets suitable for modern language models is Doplhin18K, consisting of 18 thousand math word problems (MWPs) scraped from community question-answering (CQA) math posts from the Yahoo! Answers' math section. Final answers were obtained by training Support Vector Machine (SVM) models to extract them based on multiple natural language solutions. After this, human annotators were employed to correct potential mistakes. Unfortunately, the complete dataset remains unpublished, with only a scraping script being available.

Another such dataset is AQuaRAT, containing 100 thousand multi-choice algebraic word problems. It is probably the first one to contain natural language rationales leading to a final answer. However, there is a high uniformity of tasks.

MathQA was created in an attempt to correct the flaws of AQuaRAT, using a subset of the tasks and converting a natural language rationale into a chained formula of predefined operations. The problems have been reported to have low language diversity and 27% of the tasks have been found to include inconsistencies [?].

ASDiv tries to mitigate the issue of high problem uniformity by aggregating math problems from 28 websites and filtering out redundant ones to avoid similar sentence patterns based on their proposed metric - Corpus Lexicon Diversity. Additionally, tasks are annotated with grade level and problem type. Despite being linguistically diverse, it is relatively small, containing about 2.3K problems.

GSM8K is one of the widely used math datasets, consisting of 8.5K math word problems of primary school requiring basic arithmetic operations. They were created manually by human workers, aiming at linguistic diversity and minimal errors at the

same time. Final answers are accompanied by step-by-step solutions in natural language.

In contrast to previously mentioned datasets, which primarily focus on elementary school mathematics, MATH comprises 12.5K math competition problems, offering significantly more challenging tasks. The problem complexity prompted the authors to create an auxiliary pre-training dataset AMPS, enabling models to learn fundamentals of mathematics before being tested on MATH.

SVAMP was created by a careful selection of problems from ASDiv and MAWPS. For each task, they created new tasks by varying MWP structure to evaluate whether language models really do understand the assignments and do not rely on simplistic heuristics.

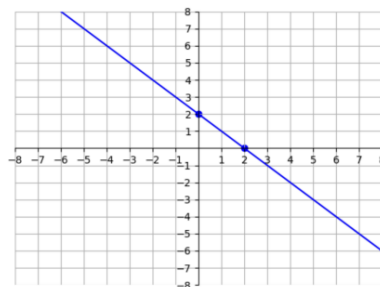
Although not explicitly stated, the aforementioned datasets were in the English language. Unfortunately, low-resource languages generally lack such benchmarks, with mathematics being no exception. However, there have been attempts to create math multilingual datasets. One such dataset is MGSM, or Multilingual Generalized Structured Mathematics dataset [2]. Derived from GSM8K, it comprises 250 problems manually translated into 10 topologically diverse languages, namely Bengali, Chinese, French, German, Japanese, Russian, Spanish, Swahili, Telugu, and Thai.

OlymMATH is new and should be more challenging. Then some MathEval, which was rejected in openreview. Omni-Math even better. Mention briefly MMLU. But then, sota LLMs have totally failed USAMO high school olympiad questions.

With the rise of multimodal language models, the evaluation has shifted towards assessing mathematical tasks with visual inputs. One of such benchmarks is **MathVista** [1]. Another one is **DynaMath**, but it seems there are only graph-based questions with multiple-seeds.

Overall, performance on math word problems seems to be a widely used metric when referring to reasoning abilities in LLMs. One of the reasons could be that other types of tasks, such as linguistic reasoning, require only a limited level of reasoning. Furthermore, in terms of informal reasoning, there is a challenge of creating tasks precise enough to avoid ambiguity and the possibility of multiple solutions. In contrast, tasks requiring rigorous formal reasoning may struggle to find practical application in the natural language domain. Hence, math word problems try to compromise between these two approaches by using well-defined problems with specific answers in natural language. Still, during the creation of such datasets, it is crucial to avoid ambiguously stated MWPs and *plug-and-chug* calculations that can be solved mechanically. Both types are unsuitable for evaluating true reasoning abilities.

Q96 from DYNAMATH: What is the distance between the x -intercept and y -intercept of blue line?



Claude-3.5 Answer:

1. Identify the x -intercept and y -intercept: The x -intercept is where the line crosses the x -axis ($y = 0$). From the graph, this appears to be at $(3, 0)$. The y -intercept is where the line crosses the y -axis ($x = 0$). From the graph, this appears to be at $(0, 4)$.

Figure 2.1: Figure reading error.

Btw, ak sme stale len niekde tu, tak to dakujem pekne. Bude to dobra sranda.

Verim, ze vo vlemi blizkej buducnosti taketo somariny niekto vyriesi :D

2.2 Evaluation

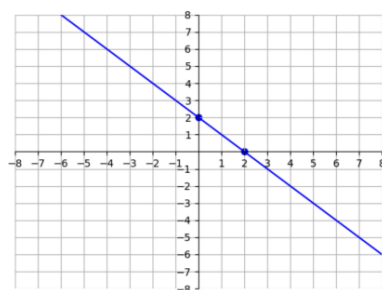
[Rule-based evaluation (multi-choice being very popular, but...) vs free-form answers (not very popular...)]

2.3 Prompting techniques

2.3.1 Chain-of-thought

2.3.2 Cross-lingual

Q96 from DYNAMATH: What is the distance between the x -intercept and y -intercept of blue line?



Claude-3.5 Answer:

1. Identify the x -intercept and y -intercept: The x -intercept is where the line crosses the x -axis ($y = 0$). **From the graph, this appears to be at $(3, 0)$.** The y -intercept is where the line crosses the y -axis ($x = 0$). **From the graph, this appears to be at $(0, 4)$.**

Figure 2.2: Error classification objection.

Niekedy nie uplne suhlasim s pridelenou error class. A naozaj je niekedy tazke povedat, kde nastava chyba. Napríklad tuto by som ja povedala, ze model len nevie "citát" ten obrazok. Asi si zle pospajal, co je co a na co pozerat (ale to je len moj pocit, ze?). To by ma tiez zaujimalo, ze ako dobre vedia prepajat modely medzi objektmi na obrazku a v texte. Napr. "uvedomuje si", co je vyska, a ako je znazornena jej dlzka na obrazku? Alebo vobec dlzku ktorych stran znazornuju tie cisla na obrazku?

Chapter 3

Dataset

The prevailing number of mathematical benchmarks uses only textual descriptions and only a few introduce visual contexts. This disparity makes the reasoning abilities of visual-language models underexplored, especially in low-resource languages like Slovak. To bridge the gap, we introduce a new curated benchmark for assessing visual and mathematical reasoning abilities of multimodal models.

The dataset comprises [tba] textual questions accompanied by images from standardized mathematical exams, namely [tba].

3.1 Data collection

This section describes the data collection process. We define the criteria of high-quality visual math reasoning tasks, describe the sources from which they were obtained, and outline the taxonomy with the annotation of our dataset. The final dataset examples can be seen in Figure [tba].

3.1.1 Criteria

[What makes for us a good reasoning math task.]

3.1.2 Sources

We collect questions from standardized math exams that are held every year in Slovakia. These sources represent a wide range of students, grade levels, difficulty, and thus ensure a diverse coverage of mathematical reasoning tasks.

Matura (2000–2025)

The Matura exam is the Slovak high school graduation exam. The math section targets students typically aged 18–19 and covers a wide spectrum of topics including algebra,

functions, geometry, statistics, and word problems. The problems require conceptual understanding, procedural knowledge, and occasionally also interpretation of charts and diagrams. We collected a total of [tba] word problems from this source.

Monitor (2005–2025)

The Monitor exam is a national standardized test taken by ninth-grade students (around age 14). The math section includes word problems with arithmetic, basic algebra, geometry, and real-world applications, generally framed in a non-technical language to assess practical mathematical reasoning. We collected [tba] tasks from this source.

Testovanie 5 (2013–2022)

Testovanie 5 is a national standardized test administered to fifth-grade students (around age 11) in elementary school. The exam evaluates basic numeracy, arithmetic operations, geometry, and simple word problems relevant to everyday contexts. The problems are designed to be solvable without advanced symbolic manipulation and emphasize basic reasoning and comprehension. From this source, we gathered [tba] problems.

Mathematical Olympiad

[Here I am not sure how many problems...] We also included selected problems from various levels of the Mathematical Olympiad, a nationwide math competition held annually for students across different age groups. The tasks are designed to challenge the best-performing students and are categorized by age (Z5–Z9 for grades 5 to 9, and SŠ for secondary school levels). Olympiad problems are often non-standard and require deeper mathematical insight, multi-step reasoning, and sometimes visual or geometric intuition. We extracted [tba] problems from this source.

3.1.3 Taxonomy

Image type

- plot
- diagram
- geometry object
- planar object
- solid object
- other a

3.1.4 Annotation

[Describe the annotation process if needed]

3.1.5 Questions

- What about multi-choice questions, where one choice = one image?

3.2 Data analysis

Chapter 4

Experiments

The chapter describes our evaluation methodology for assessing the visual mathematical reasoning abilities of VLMs on our newly curated benchmark. As the first step, we evaluate [tba] models in total under several prompting techniques. [tba]

4.1 Models

We evaluate [tba] models, [tba] of which are open-source, namely...

- Phi-4 Multimodal Instruct
- Llama 4 Maverick
- GPT-4v?

4.2 Methods

- baseline
- zero-shot cot
- few-shot cot
- cross-lingual

4.3 Evaluation

Chapter 5

Results

In this chapter, we present the results of our experiments.

5.1 Limitations

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